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Mathematics and Statistics

Introduction to linear algebra MAT 1341 A
Test 1 – version A

11 February, 2025. 13:00

Instructions:

- The length of this test is **75 minutes**.
- Question 1 consists of short-answer questions. You do not need to justify your answers in this question.
- Questions 2 to 4 are long-answer questions. **To get full marks on these questions, you must justify your answers, and your solutions must be clear, logical, and readable.**
- Question 5 is a bonus question with 3 points. To get marks on this question, your solution must be completely correct. You should not work on this question before you have finished all the others.
- There is a blank page at the end of the test. If you need extra space for any question, please use the blank page and/or the backs of pages. Indicate clearly where your answer can be found.
- **Cellular phones, smart watches, unauthorized electronic devices, or course notes are not allowed during this exam. Phones and devices must be turned off and put away in your bag. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.**

  By signing below, you acknowledge that you have ensured that you are complying with the above statement.
- Good luck!

Last name :_____ First name :_____

Student number: _____ Signature :_____

Q1	Q2	Q3	Q4	B	Total

(1) Determine whether each of the following statements is true or false.

(a) A homogeneous linear system with 500 equations and 500 variables always has a unique solution.

(b) For any $a, b, c, d, e, f \in \mathbb{R}$, the matrices

$$\begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} a+c & b+d \\ a+e & b+f \\ a & b \end{bmatrix}$$

have the same rank.

(c) If $A, B, C \in M_{3\times 3}(\mathbb{R})$ satisfy

$$AB = AC$$

then either $B = C$ or A is the zero matrix.

(d) If V is a vector space, W is a subspace, and $w_1, w_2, w_3 \in W$, then $W \subseteq \mathrm{Span}\{w_1, w_2, w_3\}$.

(e) $\{A \in M_2(\mathbb{R}) : A = A^T\} = \mathrm{Span}\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\}$.

(f) If $A, B \in M_{4\times 4}(\mathbb{R})$ then $(AB)^T = B^T A^T$.

Below, circle T for statements that are true and F for statements that are false. [6]

(a)	(b)	(c)	(d)	(e)	(f)
T/F	T/F	T/F	T/F	T/F	T/F

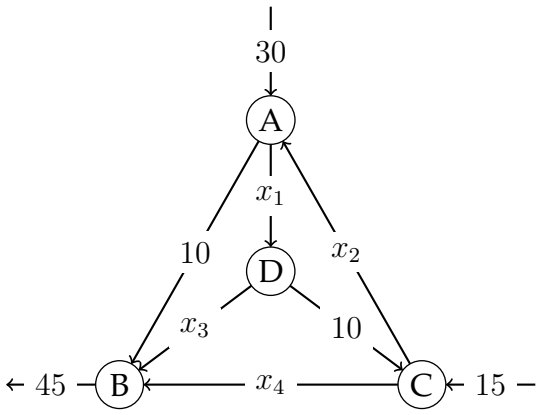
Solution: F T F F T T

Let $A := \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$. Determine A^{2025}. You do not need to justify your answer. [4]

Answer: _____

Solution: $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2025 & 0 \end{bmatrix}$.

(2) Below is a network of one-way streets:



(a) Find the general solution for the flow. [6]

Solution: The network gives the linear system

$$\begin{array}{rrcl} \text{A}: & x_1 + 10 & = & x_2 + 30 \\ \text{B}: & 45 & = & x_3 + x_4 + 10 \\ \text{C}: & x_2 + x_4 & = & 10 + 10 \\ \text{D}: & x_3 + 10 & = & x_1 \end{array}$$

with augmented matrix

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 20 \\ 0 & 0 & -1 & -1 & -35 \\ 0 & 1 & 0 & 1 & 25 \\ -1 & 0 & 1 & 0 & -10 \end{array}\right].$$

We solve this by Gaussian elimination:

$$\begin{array}{r} -R_4 \to \\ R_3 \to \\ -R_2 \to \\ R_1 + R_4 \to \end{array} \left[\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 10 \\ 0 & 1 & 0 & 1 & 25 \\ 0 & 0 & 1 & 1 & 35 \\ 0 & -1 & 1 & 0 & 10 \end{array}\right] \qquad \begin{array}{r} R_1 + R_3 \to \\ \\ \\ R_4 + R_2 \to \end{array} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 45 \\ 0 & 1 & 0 & 1 & 25 \\ 0 & 0 & 1 & 1 & 35 \\ 0 & 0 & 1 & 1 & 35 \end{array}\right]$$

$$\begin{array}{r} \\ \\ \\ R_4 - R_3 \to \end{array} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 45 \\ 0 & 1 & 0 & 1 & 25 \\ 0 & 0 & 1 & 1 & 35 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

We use a parameter $x_4 = t$ and then get $x_1 = 45 - t$, $x_2 = 25 - t$, $x_3 = 35 - t$.

$x_i \geq 0$ for $i = 1, \ldots, 4$, and so $0 \leq t \leq 25$.

(b) What is the maximum flow along the route AD? [2]

Solution: $x_1 = 45 - t$ with $0 \leq t \leq 25$.

Hence the maximum is 45.

(c) If the route AC is closed, will there be a traffic jam? [2]

Solution: This would mean $x_2 = 0$. Using $x_2 = 25 - t$, we get $t = 25$.

The system still has a solution ($x_1 = 20, x_3 = 10, x_4 = 25$) so there is no traffic jam.

(3) (a) Determine if the set $S = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ is a subspace of $\mathbb{R}^2$. Justify your answer. [5]

Solution: $(0, 0) \notin S$ since $0^2 + 0^2 = 0 \neq 1$.

Therefore, S is not a subspace.

(b) Let T be the general solution to a **homogeneous** linear system with n variables. Show that T is a subspace of $\mathbb{R}^n$. [5]

Solution: To reduce notation let us write the system using matrix multiplication, as $Ax = \vec{0}$, where $A \in M_{m\times n}(\mathbb{R})$ and

$$x = \begin{bmatrix} x_1 \\ \cdots \\ x_n \end{bmatrix}.$$

Then $T = \{x \in \mathbb{R}^n : Ax = \vec{0}\}$.

We'll use the Subspace Test.

(1): $A\vec{0} = \vec{0}$, so $\vec{0} \in T$.

(2): If $x, y \in T$ then $Ax = \vec{0}$ and $Ay = \vec{0}$. Hence $A(x + y) = Ax + Ay = \vec{0} + \vec{0} = \vec{0}$. This shows that $x + y \in T$.

(3): If $x \in T$ and $c \in \mathbb{R}$ then $Ax\vec{0}$. Hence $A(cx) = cAx = c\vec{0} = \vec{0}$. This shows that $cx \in T$.

Therefore by the Subspace Test, T is a subspace.

(4) (a) Is the vector $(1,2,3)$ in $\mathrm{Span}\{(-1,0,1),(1,1,1)\}$? Justify your answer. [5]

Solution: Yes, since $(1,2,3) = 1(-1,0,1) + 2(1,1,1)$.

(One can find this expression by solving the linear system $a_1(-1,0,1) + a_2(1,1,1) = (1,2,3)$, for a_1, a_2.)

(b) Find an equation which describes the plane $\mathrm{Span}\{(-1,0,1),(1,1,1)\}$ in $\mathbb{R}^3$. [5]

Solution: The plane corresponds to (x,y,z) such that there is a solution to $a_1(-1,0,1)+a_2(1,1,1) = (x,y,z)$. The augmented matrix for this system is

$$\left[\begin{array}{cc|c} -1 & 1 & x \\ 0 & 1 & y \\ 1 & 1 & z \end{array}\right].$$

To characterize when it has solutions, use Gaussian elimination:

$$R_3 + R_1 \to \left[\begin{array}{cc|c} -1 & 1 & x \\ 0 & 1 & y \\ 0 & 2 & x+z \end{array}\right] \qquad R_3 - 2R_2 \to \left[\begin{array}{cc|c} -1 & 1 & x \\ 0 & 1 & y \\ 0 & 0 & x-2y+z \end{array}\right].$$

This has a solution if and only if $x - 2y + z = 0$. Hence the equation for this plane is

$$x - 2y + z = 0.$$

Alternative solution: A normal vector is given by $n = (-1,0,1)\times(1,1,1) = (-1,2,-1)$. Since the plane passes through $(0,0,0)$, it has the equation $n.v = 0$, which is

$$-x + 2y - z = 0.$$

Bonus: Suppose that $A \in M_{n\times n}(\mathbb{R})$ and $b \in \mathbb{R}^n$, and that the linear system with augmented matrix $[A|b]$ has a unique solution. Show that for any other $c \in \mathbb{R}^n$, the linear system with augmented matrix $[A|c]$ also has a unique solution.

Solution: Let $[A'|b']$ be the RREF of $[A|b]$. Then since we have a unique solution, we must have a leading 1 in every row and column of A', i.e., $A' = I$. It follows that for any $c \in \mathbb{R}^n$, the RREF of $[A|c]$ is $[I|c]$, so this system also has a unique solution.